Q1. What is the difference between Global and Local Scope of a Variable?

Ans:

In Python, the scope of a variable means the area of the program where a variable can be accessed or used.

There are two types of variable scopes:

1. Local Variable:

- A variable declared inside a function is called a local variable.
- It can be accessed only inside that function.
- It cannot be directly accessed outside the function.

Example:

```
def show():
    x = 10
    print(x)


show()
```

Output:

10

2. Global Variable:

- A variable declared outside a function is called a global variable.
- It can be accessed throughout the program, including inside functions.
- To modify a global variable inside a function, we use the global keyword.

Example:

```
x = 20


def show():
    print(x)


show()
print(x)
```

Output:

20

20

Difference between Global and Local Variables:

Local Variable	**Global Variable**
Declared inside a function.	Declared outside a function.
Accessible only inside the function.	Accessible throughout the program.
Created when the function is called.	Generally available throughout program execution.
Cannot be directly accessed outside the function.	Can be accessed inside and outside functions.

Q2. What is a Recursive Function?

Ans:

A recursive function is a function that calls itself repeatedly to solve a problem.

It continues calling itself until a condition called the base condition is satisfied. The base condition stops the recursion.

Example:

```
def factorial(n):
    if n == 1:
        return 1
    else:
        return n * factorial(n-1)


print(factorial(5))
```

Output:

120

Explanation:

- factorial() is the recursive function.
- The function calls itself using factorial(n-1).
- if n == 1 is the base condition that stops recursion.
- It calculates 5 × 4 × 3 × 2 × 1 = 120.

Real-life example: Imagine standing between two mirrors. You see your reflection repeated inside another reflection. Similarly, a recursive function keeps calling itself until the stopping condition is reached.

Remember:

- Global = Declared outside a function.
- Local = Declared inside a function.
- Recursive = A function that calls itself.